An Introduction to 3D Printing

Part 2: Practical Application

Examples, Critical Thinking, and Design

Outline

Part 1: Printers

Process

Limitations

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

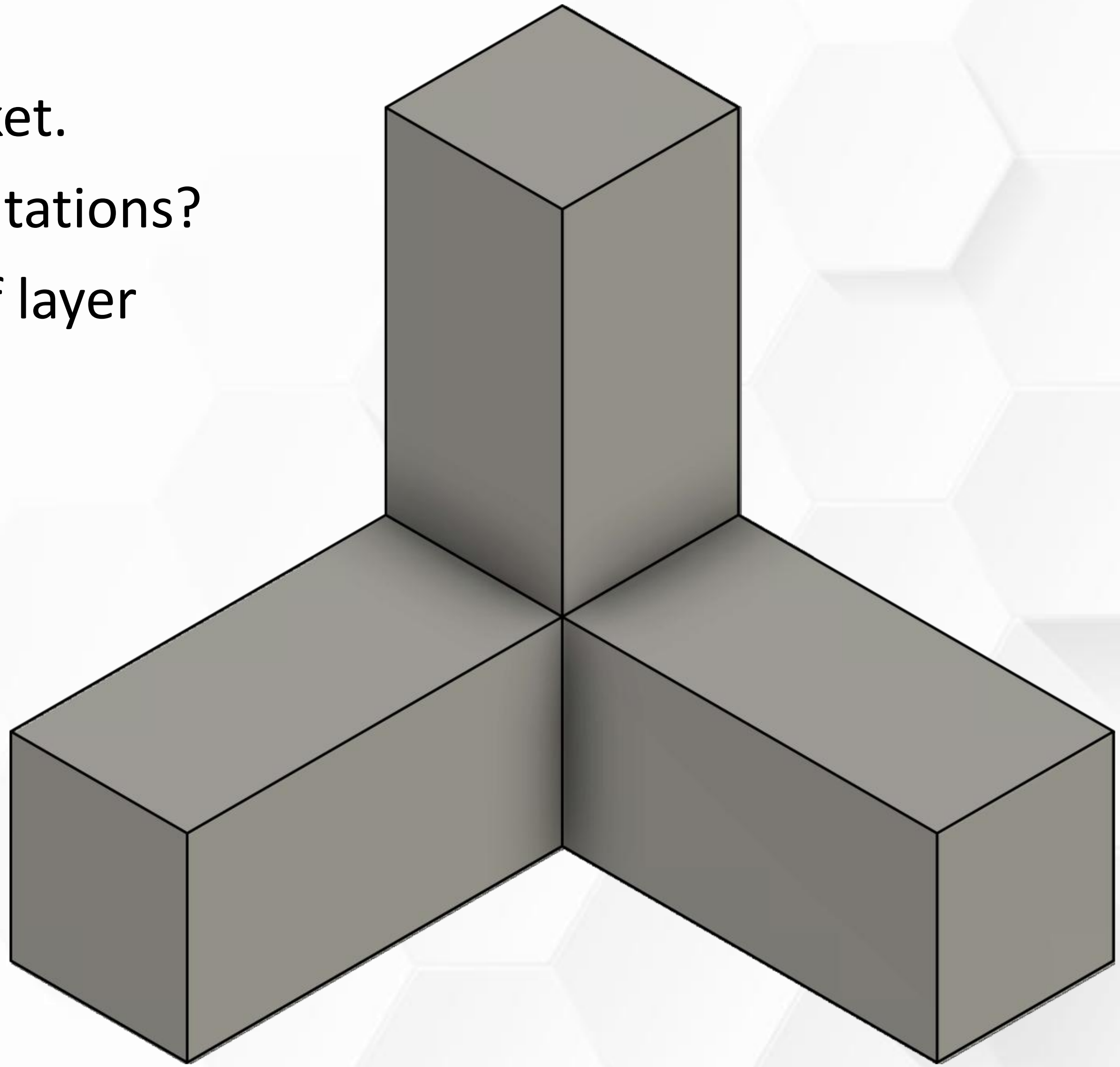
Thermal Expansion

Critical Thinking: Corners

- Rounding corners with fillets is helpful. Why?
- Stresses concentrate in corners.
  - Adding fillets improves strength.
  - Fillets reduce first layer warping.
- Print faster with more accurate movements. Why?
  - Think of a racecar (toolhead) going around a track.
  - Smoothing changes of direction reduces vibrations.

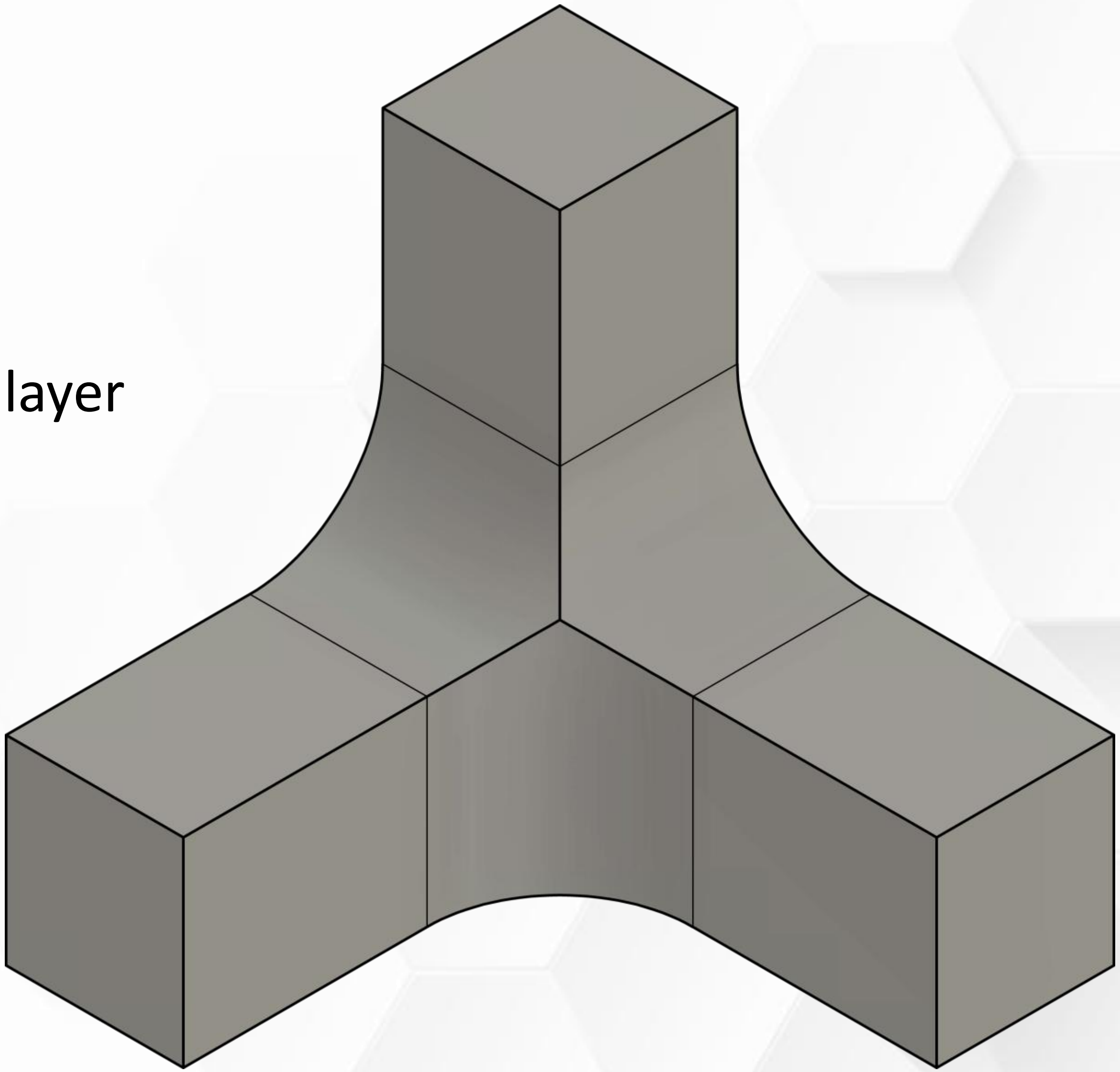
Critical Thinking: Corner Bracket

- Consider this 3-way corner bracket.
- What are the difficulties and limitations?
- Top section is weaker because of layer stacking.
- What can you do?
  - Stronger filament.
  - Stronger slicer settings.
  - What else?



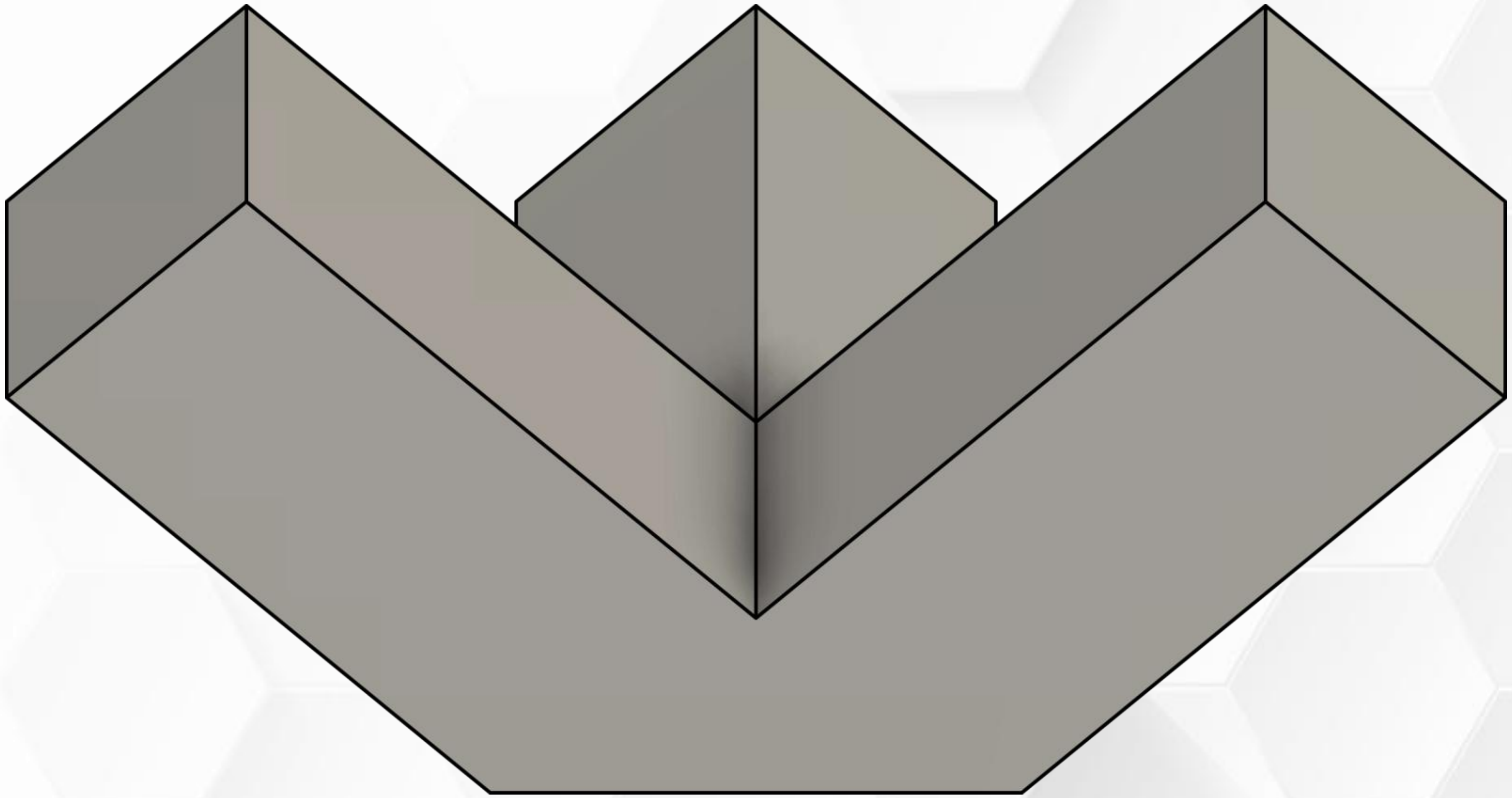
Critical Thinking: Corner Bracket

- Reinforce joints with fillets.
- Reduces stress concentration.
- Shifts the weak point upwards.
- Top section still relies on vertical layer stacking.



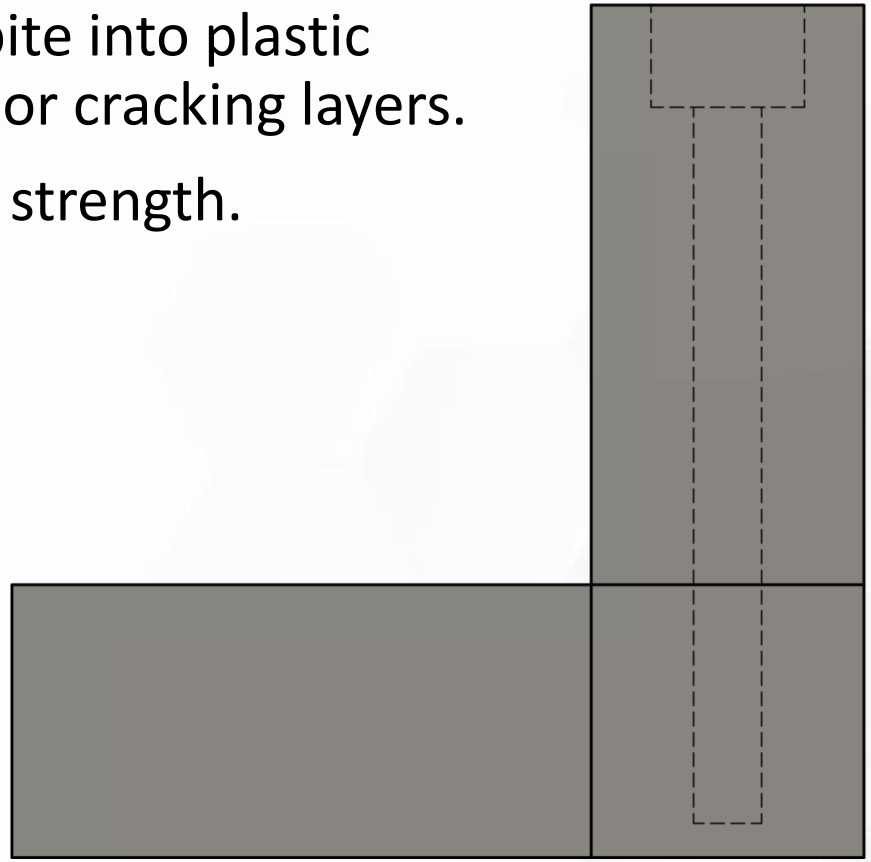
Critical Thinking: Corner Bracket

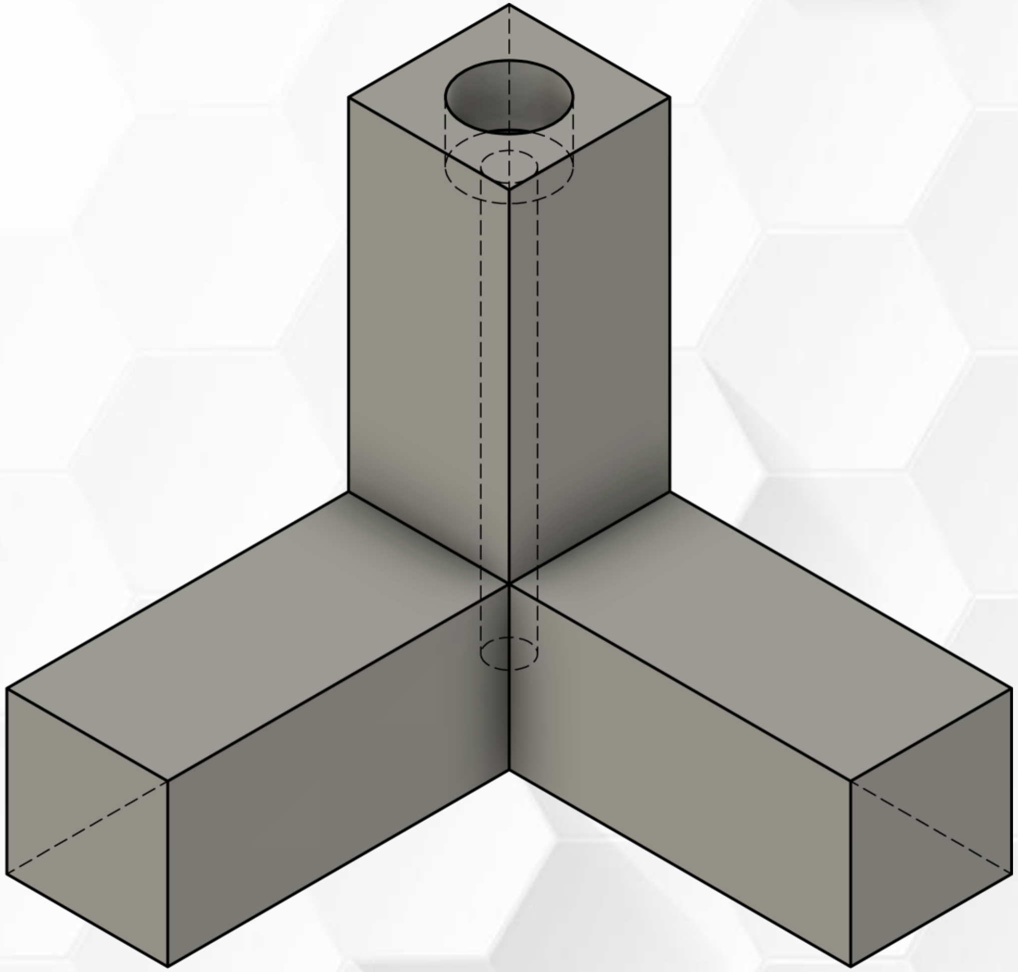
- Trim the corner to create an angled print orientation.
- This adds overhangs.
- Reduced first layer contact.
- Each section has equal strength.
- Slightly improved strength versus stacking vertical layers.



Critical Thinking: Corner Bracket

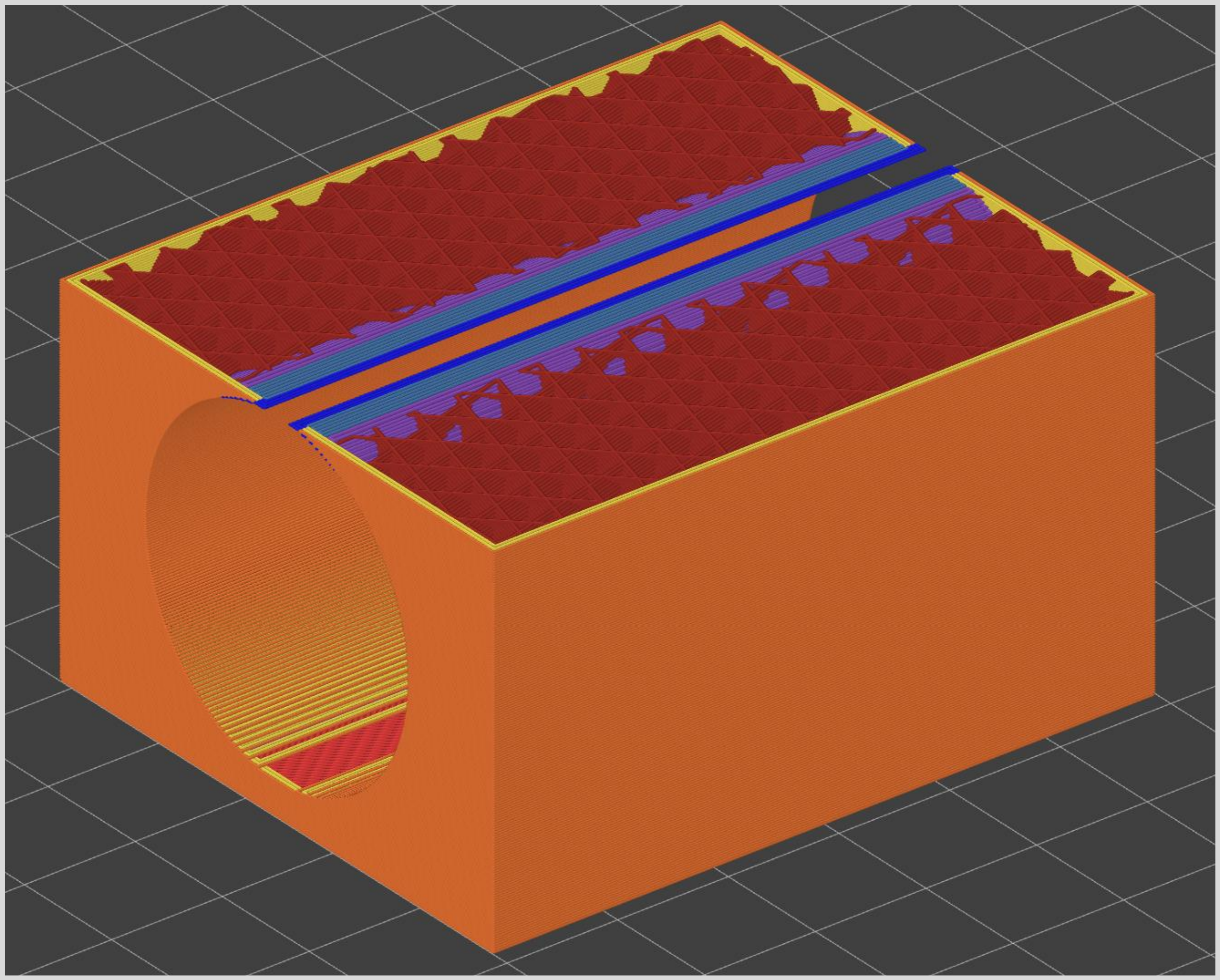
- Add tight fitting screw after printing.
- Threads should bite into plastic without splitting or cracking layers.
- Large increase in strength.





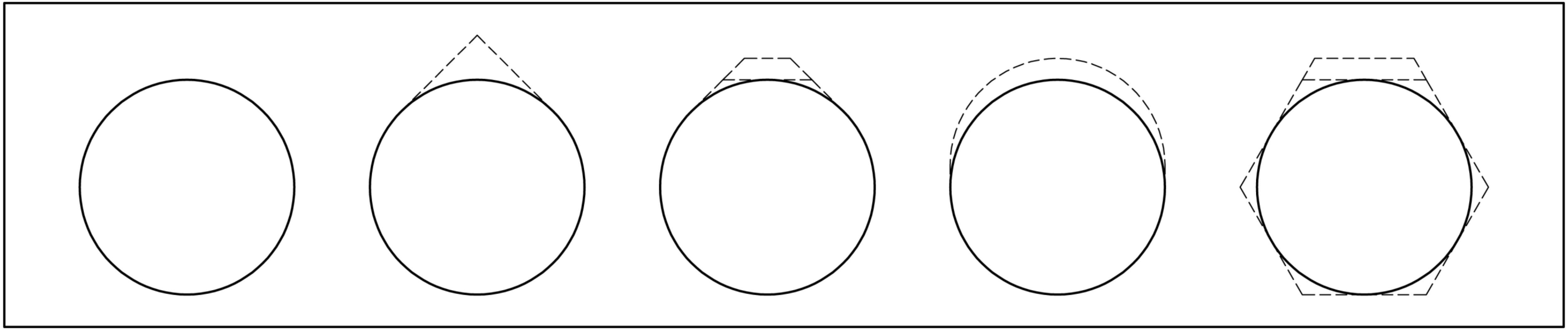
Critical Thinking: Horizontal Holes

- The top of horizontal holes have extra steep (collapsing) overhangs.
- If these extrusions sag down, the hole will be undersized.



Critical Thinking: Horizontal Holes

- How can you redesign the hole?



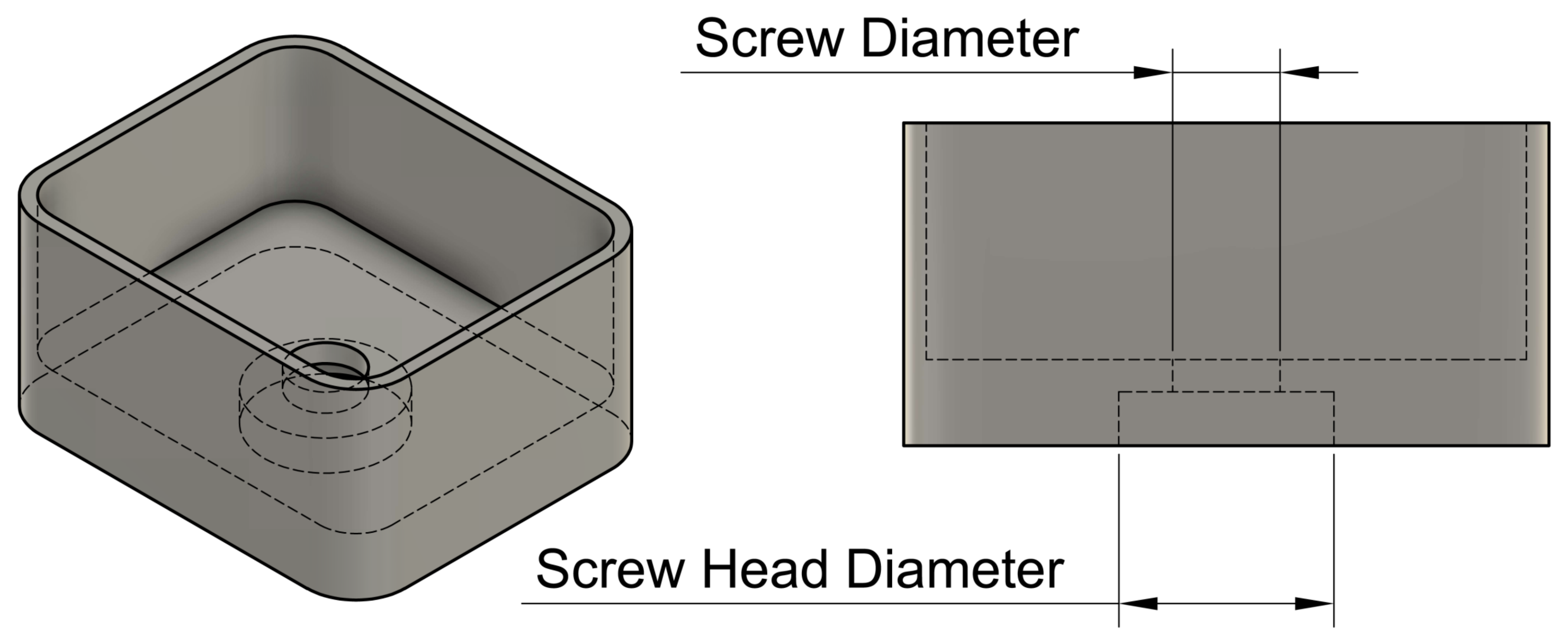
Critical Thinking: Sequential Bridges

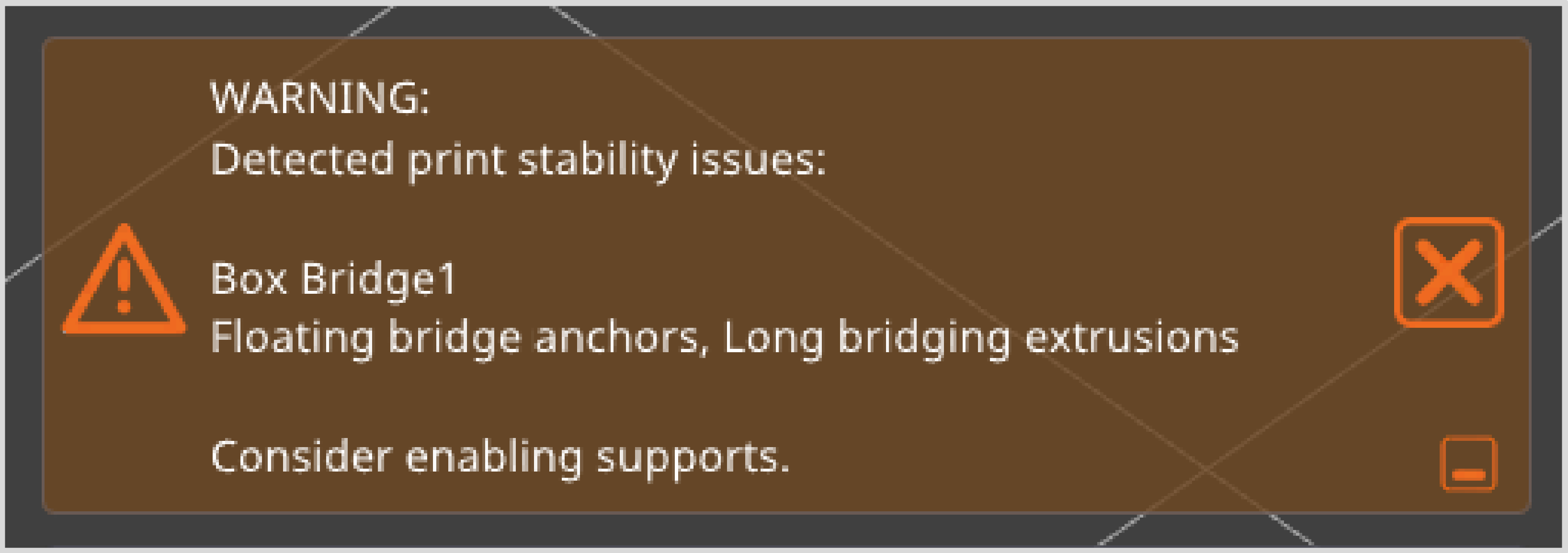
Let’s combine small bridges to create a large, multi-step bridge.

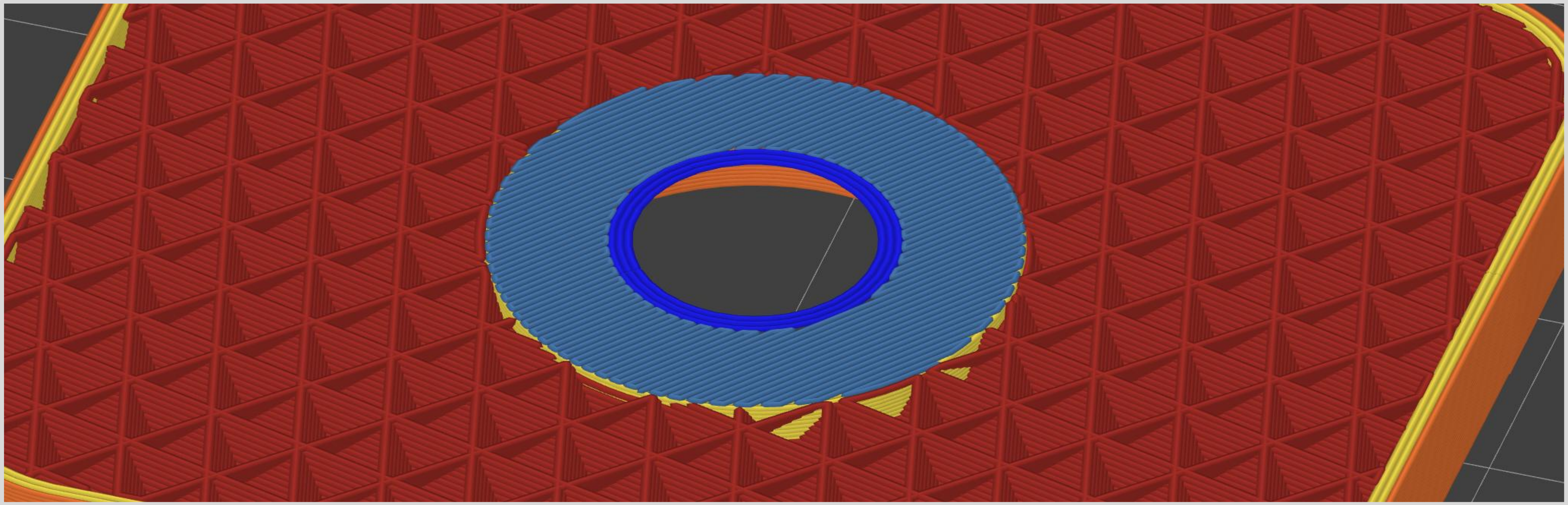
Critical Thinking: Sequential Bridges

Let's consider a storage box with a countersunk screw on the bottom.

- Can you see the hidden problem?

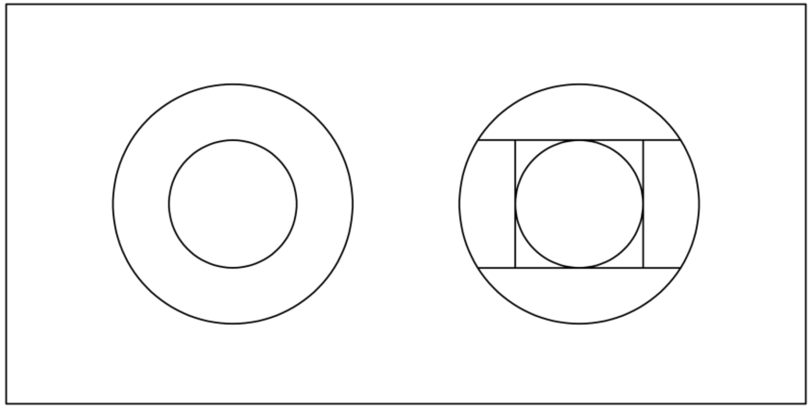


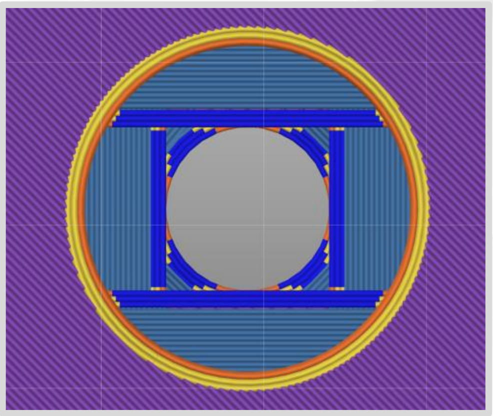

WARNING:
Detected print stability issues:
Box Bridge1
Floating bridge anchors, Long bridging extrusions
Consider enabling supports.

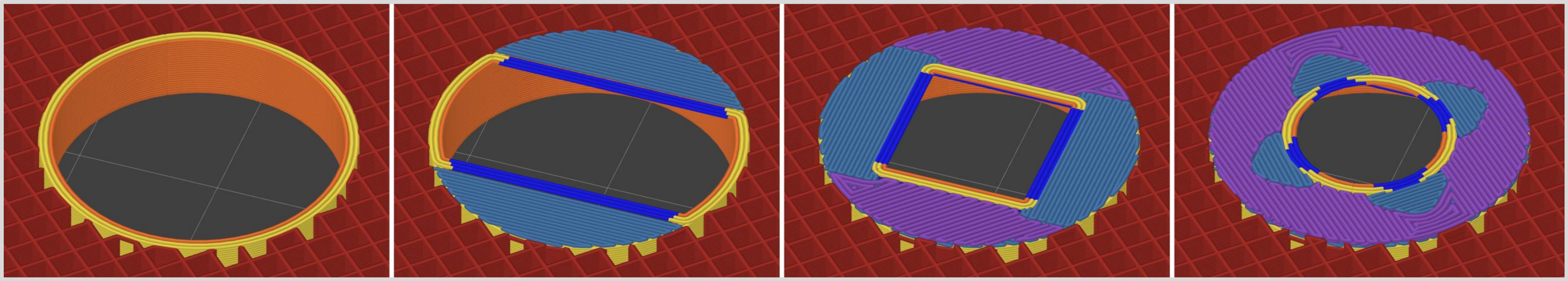


Critical Thinking: Sequential Bridges

- There are floating extrusions!
- We can fix this without supports.
- Complete the bridge in multiple steps

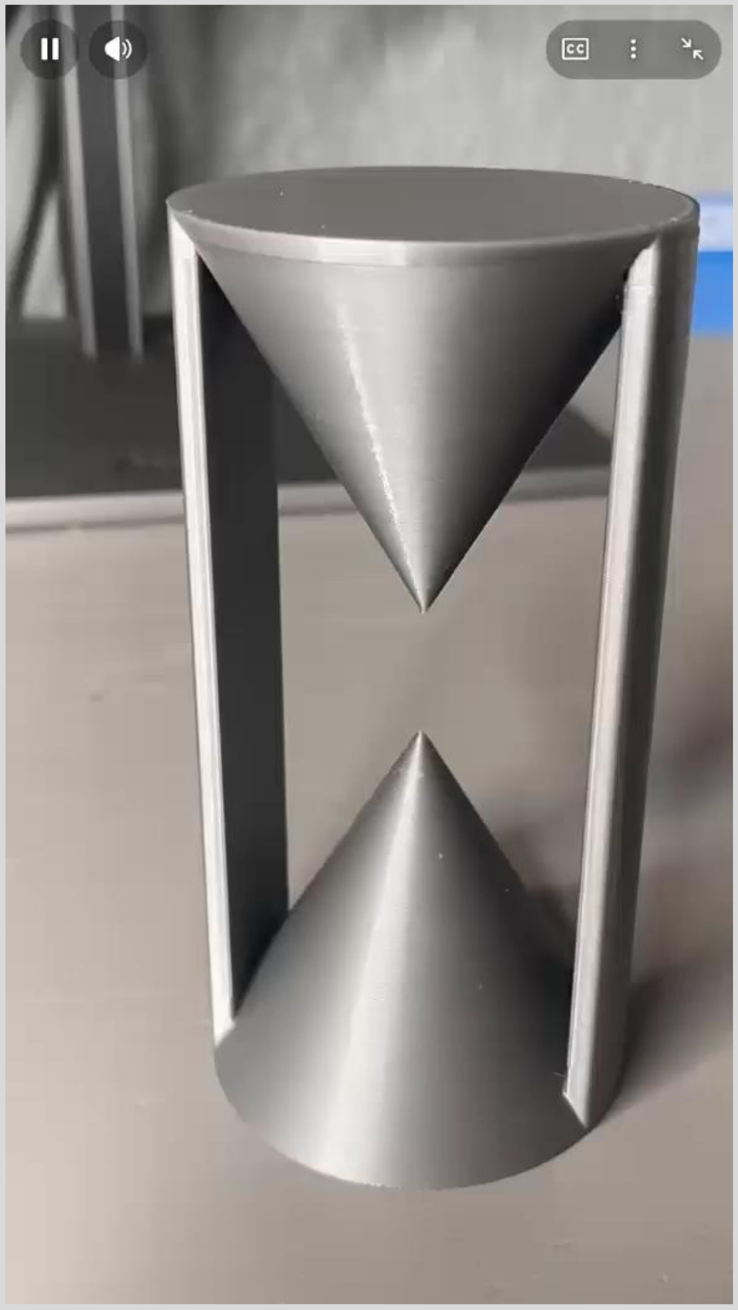






Maker Community

- I love the creativity of the maker community.
- This was printed without supports. How?
  - Put a printed part inside a printed part.

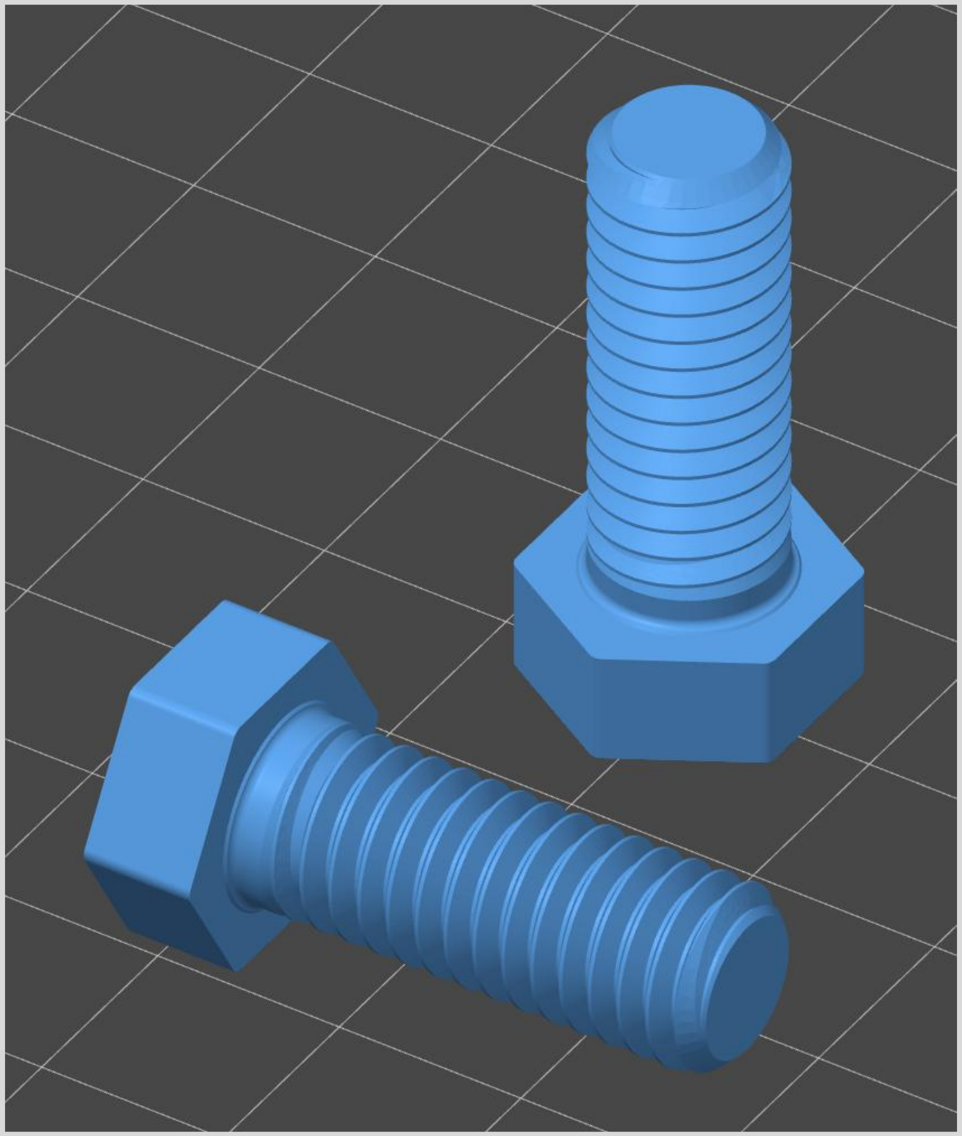


(YouTube short by christopherhelmke)

Critical Thinking: Screws

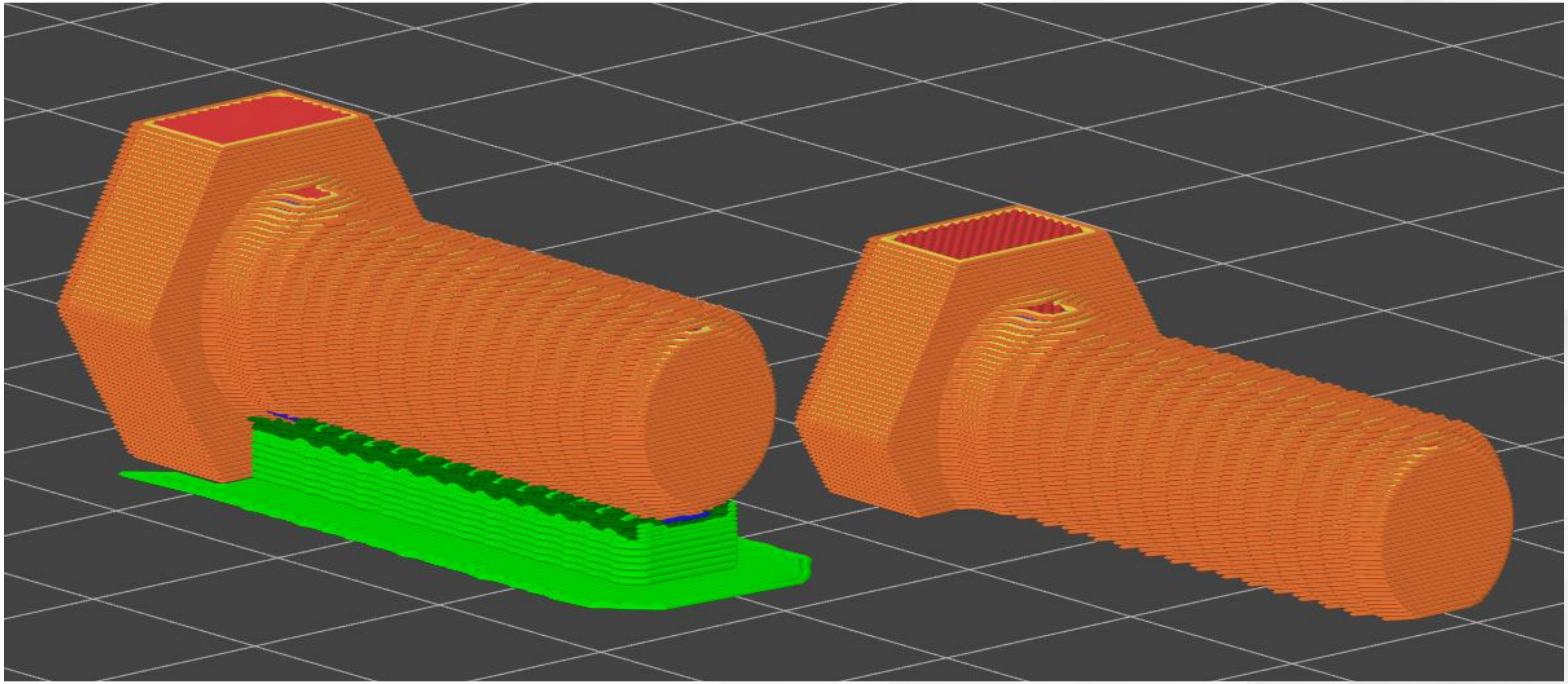
Let's 3D print a plastic screw.

- Compare and contrast print orientations.
- Horizontal.
  - Layer lines maximize strength.
  - Nozzle must be smaller than threads.
  - Bottom threads are floating extrusions.



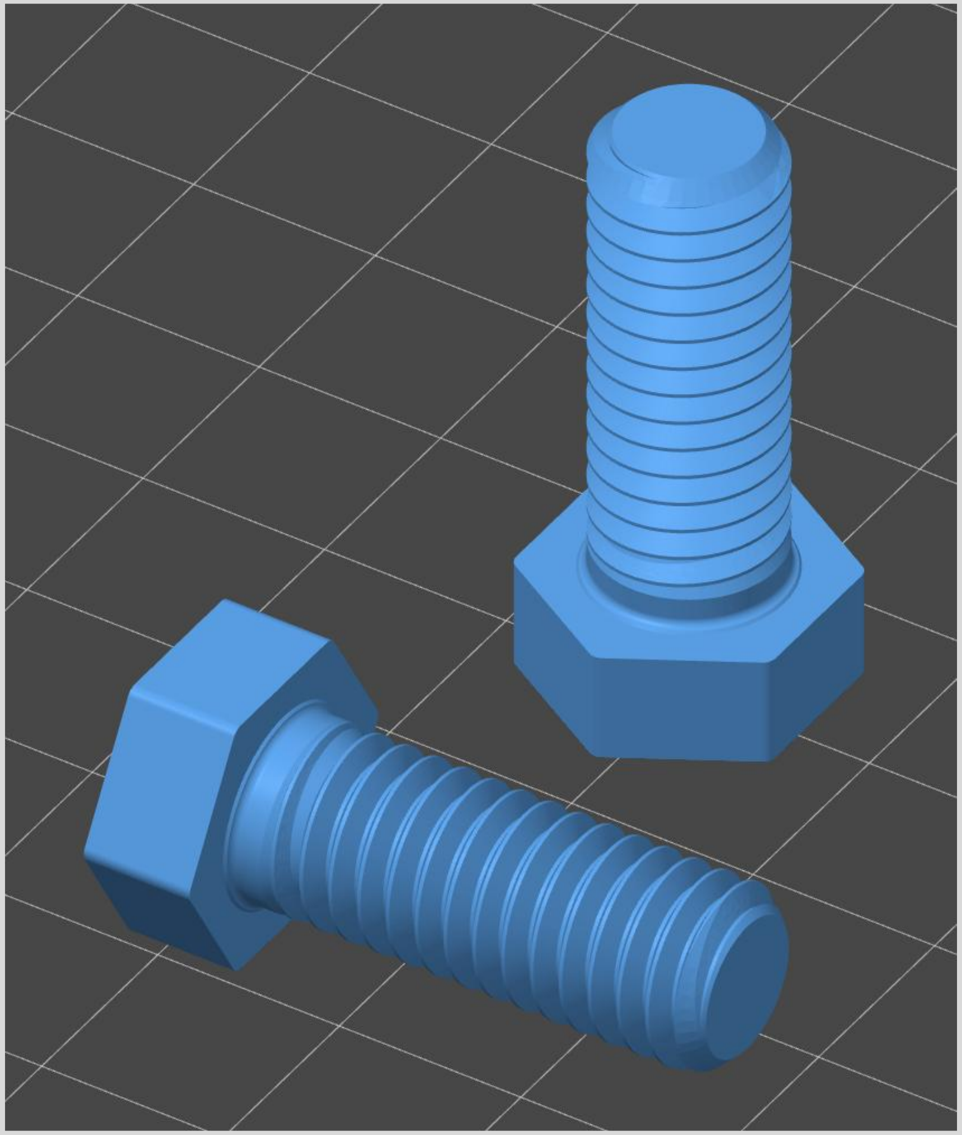
Critical Thinking: Screws

- How can you improve printability?
- Trim the bottom of the bolt so it can sit flat.



Critical Thinking: Screws

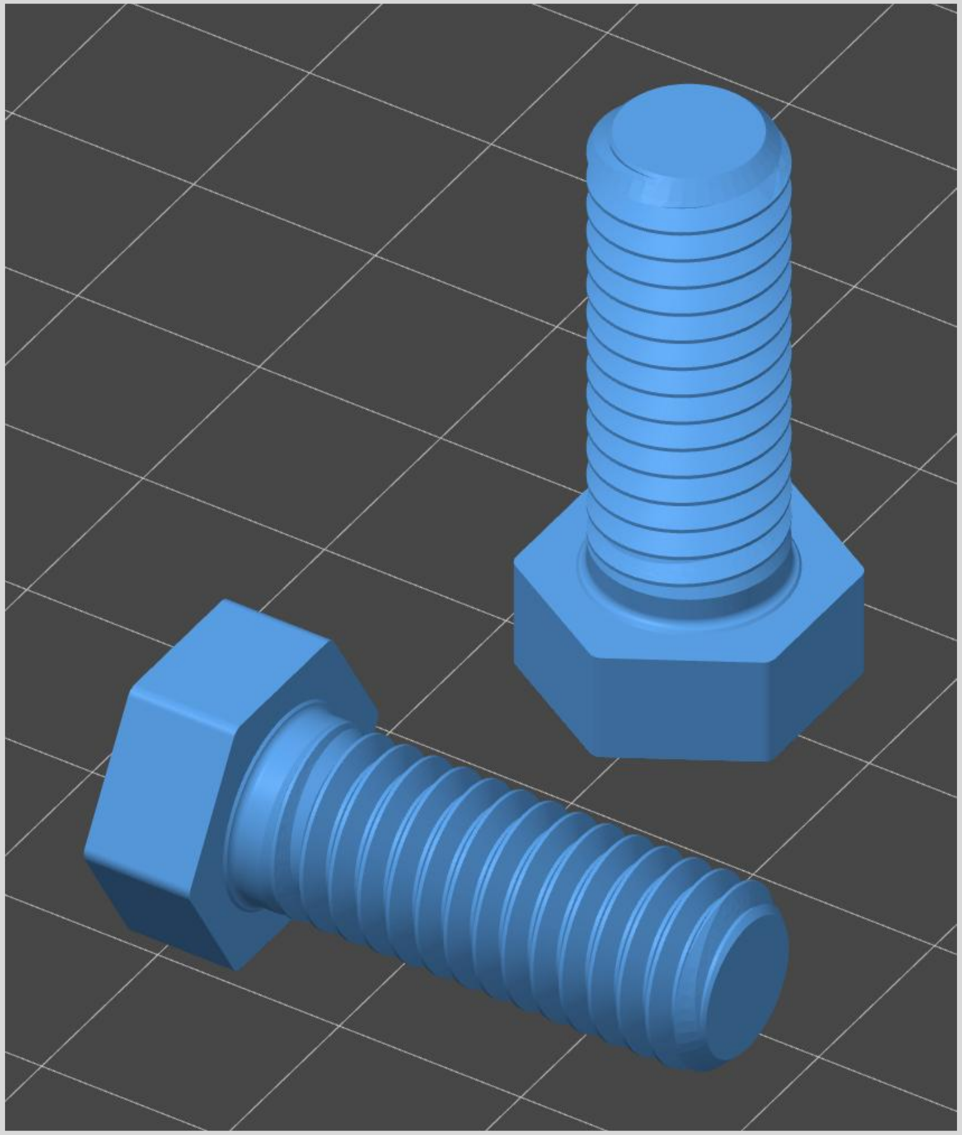
- Vertical
  - Threads limited by layer height.
  - Steep overhangs.
  - Layers lines create a weak part.
  - No supports.
  - No modifications required.
- Threads optimized for 3D printing by printables.com/@FlyingGyroscope.
- Strength testing printed bolts by My Tech Fun. And a follow-up video.



Critical Thinking: Screws

Let's summarize.

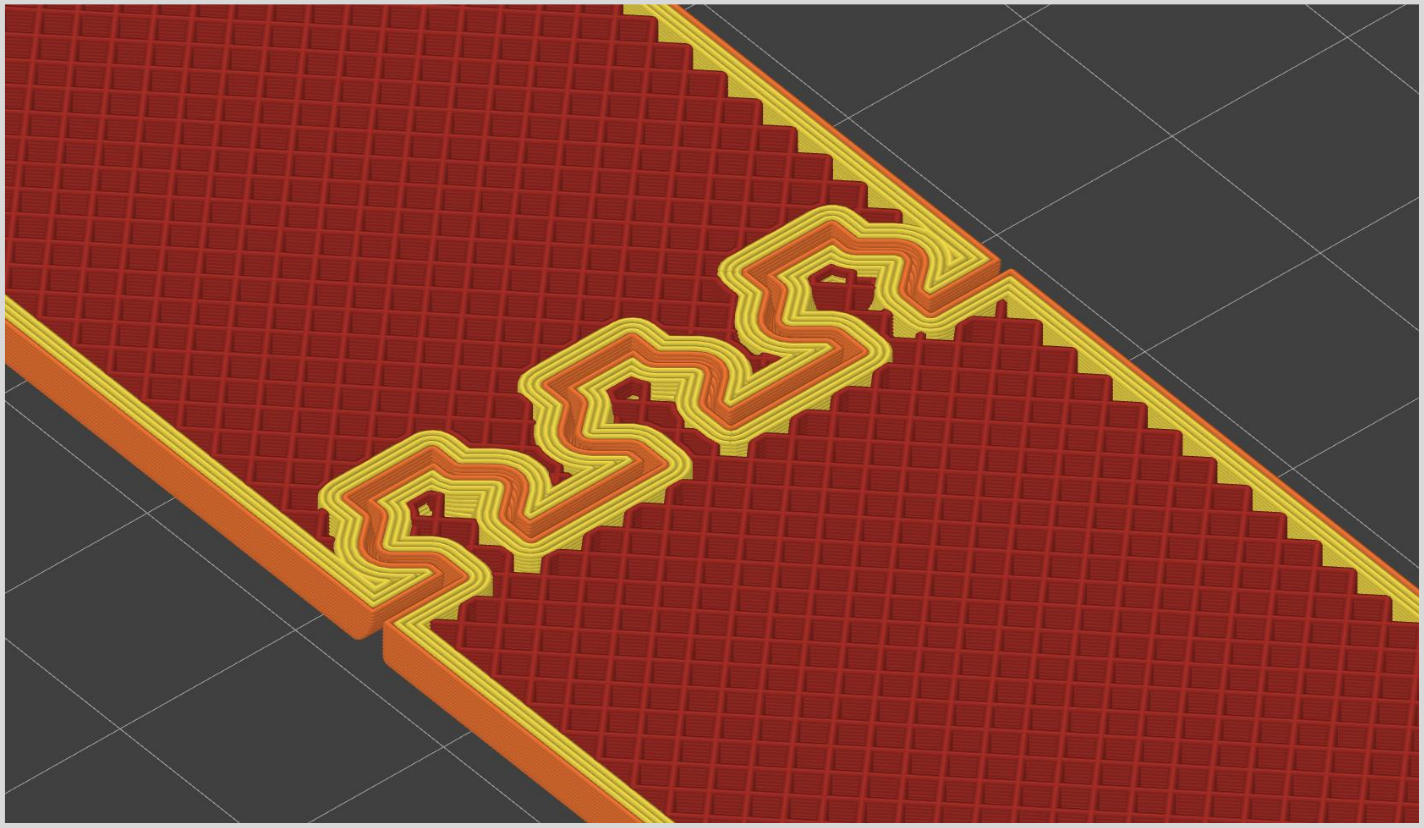
- Vertical version
  - Easy to print.
  - Looks the best.
  - Weakest.
- Horizontal version
  - Needs a minor modification.
  - Looks strange after trimming.
  - Strongest.
- These kinds of decisions and compromises are common in 3D printing.



Modeling with a Slicer

Let's make a print-in-place hinge in PrusaSlicer.

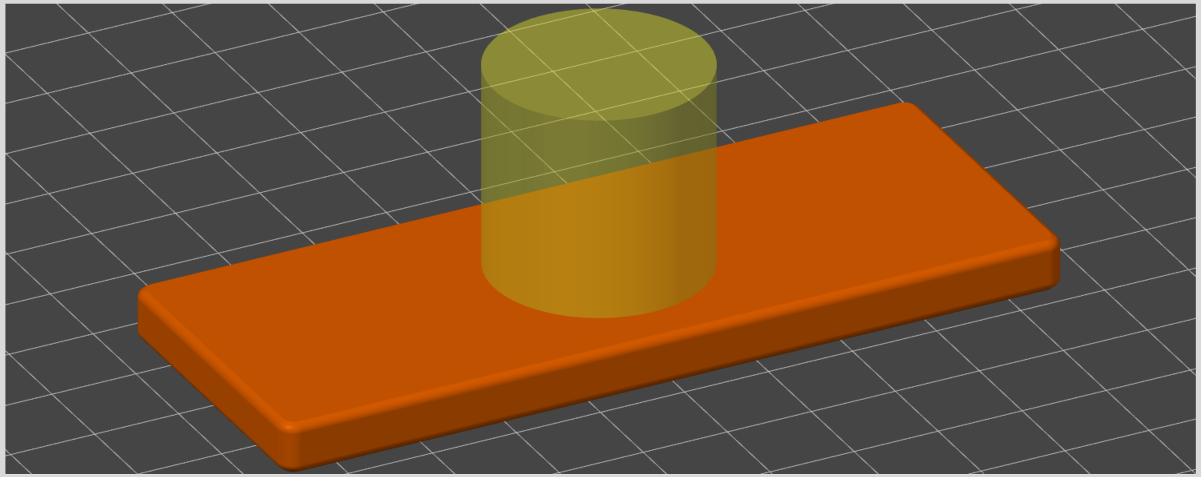
- Flexify template. Quick and easy hinges for beginners.
- Start with a flat object.
- Use negative volume to cut away material.

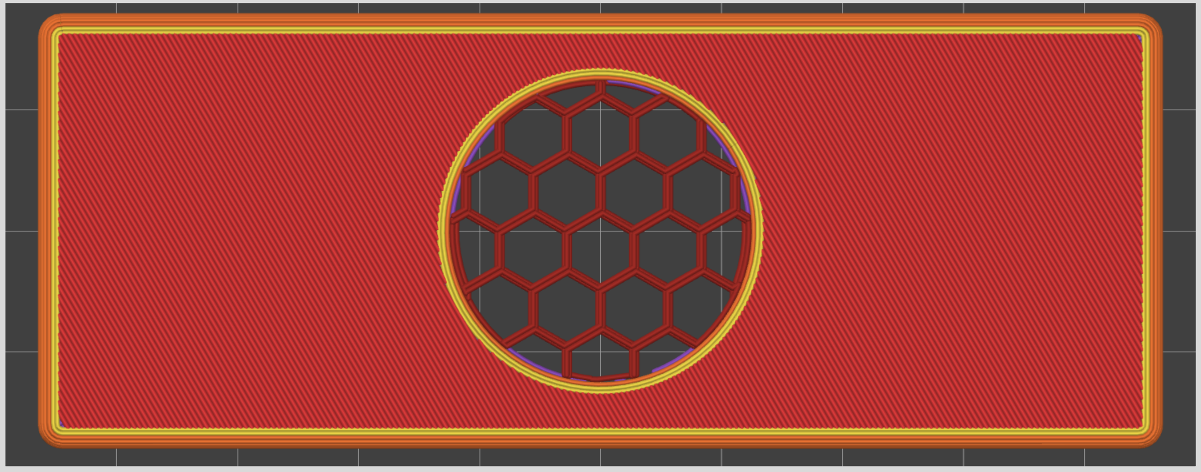


Modeling with a Slicer

Let’s make a mesh grill using PrusaSlicer.

- Define custom settings for a specific region with a **Modifier.**
- Assign zero top and bottom layers.
- Set the infill pattern.
- Change infill percent to alter the mesh size.





Critical Thinking: Snap Buckle

Let’s design a replacement part.

- Should I print a direct copy?
- What concepts and lessons from today can you apply here?
- How can you design a strong buckle?
- Compare and contrast vertical and horizontal print orientation.

(Picture from HomeDepot.)

Critical Thinking: Snap Buckle

- What choices did the designer make?

(Picture from HomeDepot.)

(Snap Buckle by Squinn.)

Critical Thinking: Snap Buckle

- Print parts laying down with horizontal print lines.
  - Good strength for the compliant mechanism.
  - Good first layer contact.
- The female socket has been reimagined.
  - Overhangs, not bridges.
  - The middle prong is a dovetail (trapezoid).

(Snap Buckle by Squinn.)


THANK YOU
imgflip.com